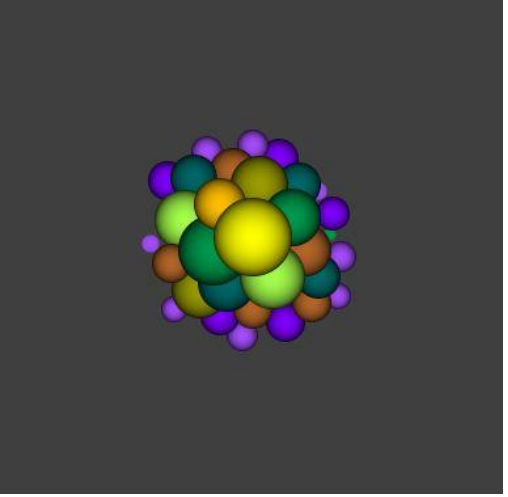
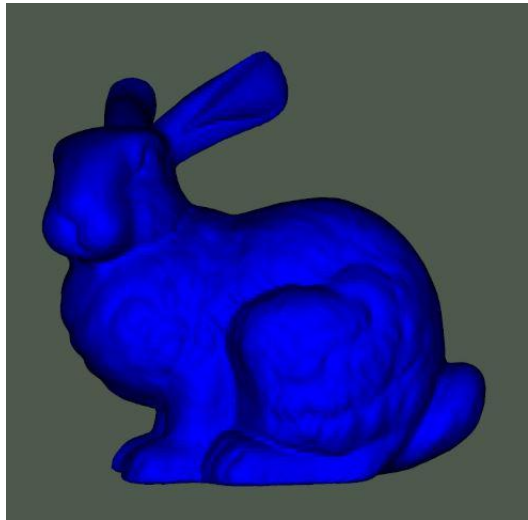


CS 440: Computer Graphics

Week 6

Lecture 12: Mesh Analysis and Operations



Acknowledgements: Cem Yuksel, University of Utah; Waqar Saleem, Habib University

Announcements and Recap

- Quiz on Monday (on Canvas)
- Assignment 1 grades will be out today (I promise)

- Recap:
  - Mesh Data Structures
    - Face Set
    - Shared Vertex
    - Adjacency Matrix
    - Winged Edge
    - Half-edge

Today: Mesh Analysis

- More about Mesh Data Structures
- Mesh Analysis
  - Subdivision
  - Smoothing
- Platonic Solids

Face Sets vs Shared Vertex

- **Face Sets:**
  - "Polygon soup"
  - List of faces
  - Unorganized
  - A vertex can appear multiple times in the same list
- **Shared Vertex:**
  - "Indexed mesh"
  - List of faces **that refer to vertices** (each vertex is given an index)
  - A little more compact than polygon soup (no repeated vertices)
  - **Face set with a shared Vertex**

Face Set vs Shared Vertex

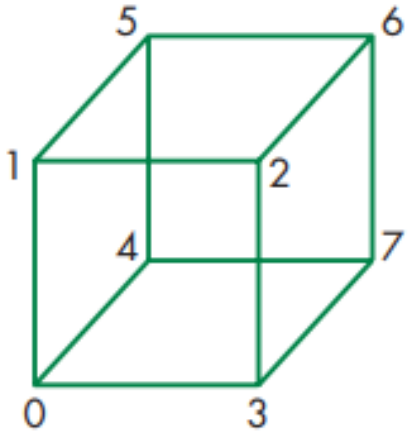
Face Set

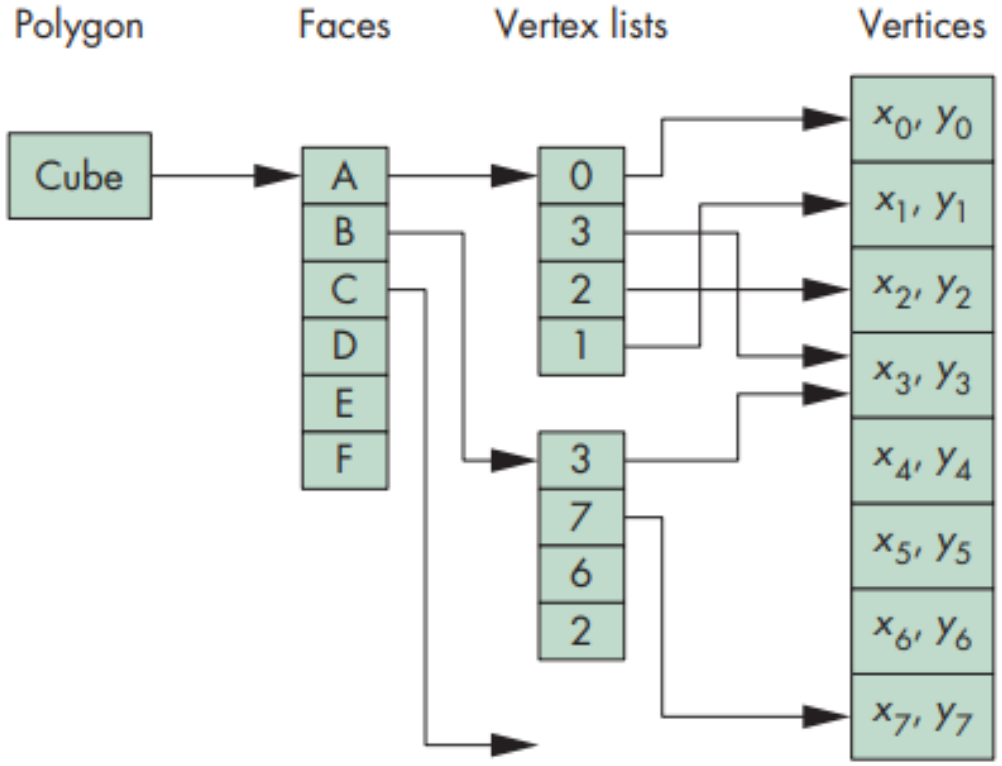
```
begin
vertex -1 -1 -1
vertex -1 1 1
vertex 1 -1 1
vertex 1 1 1
end
```

Shared Vertex

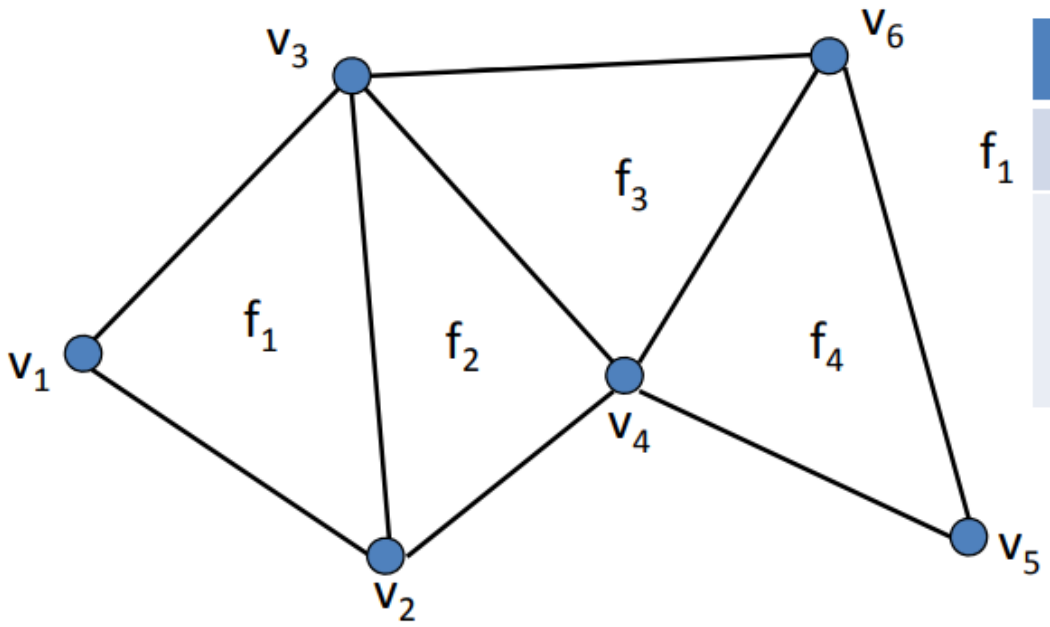
```
v -1 -1 -1   → 0
v -1 1 1     → 1
v 1 -1 1     → 2
v 1 1 1      → 3
f 0 1 2
f 0 2 3
f 0 3 1
f 2 1 3
```

Face Set with Shared Vertex





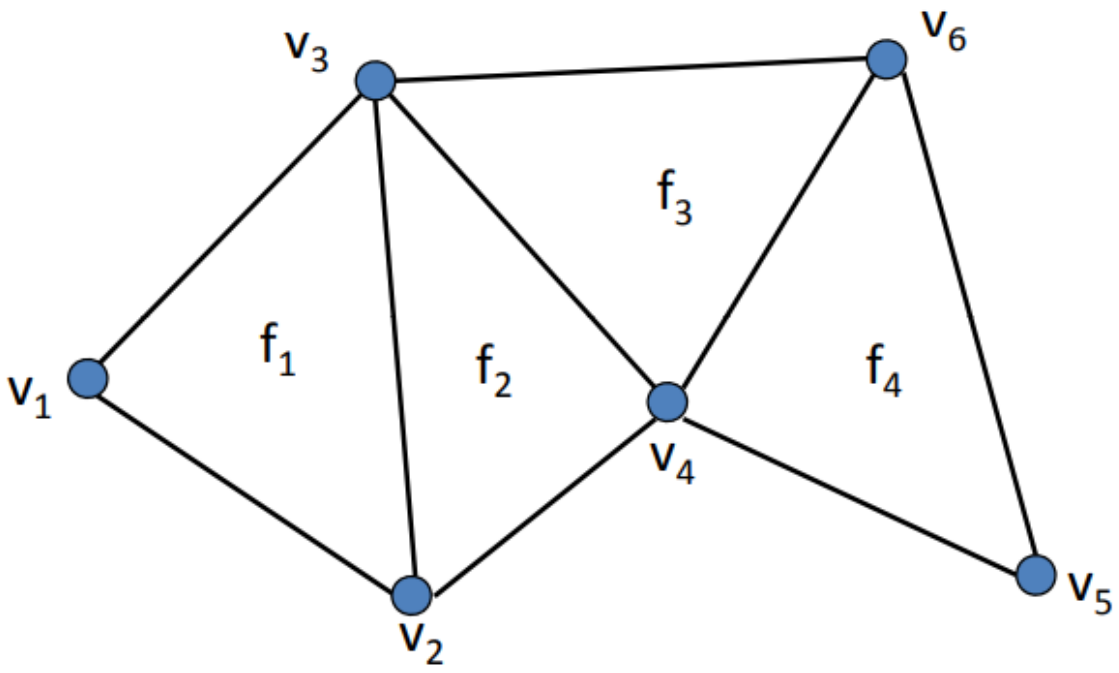
Shared Vertex



	TRIANGLES		
f_1	2	3	1
		.	
		.	
		.	

	VERTICES
v_1	[20 10 0]
v_2	[19 20 0]
v_3	[14 15 0]
	.
	.
	.

Adjacency Matrix

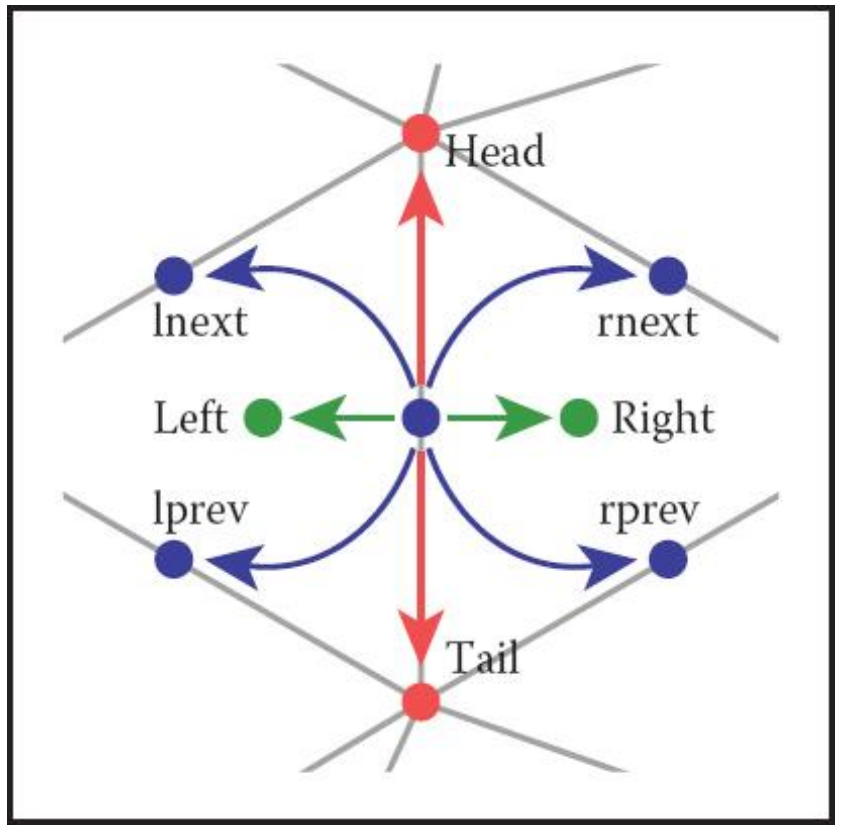


A =

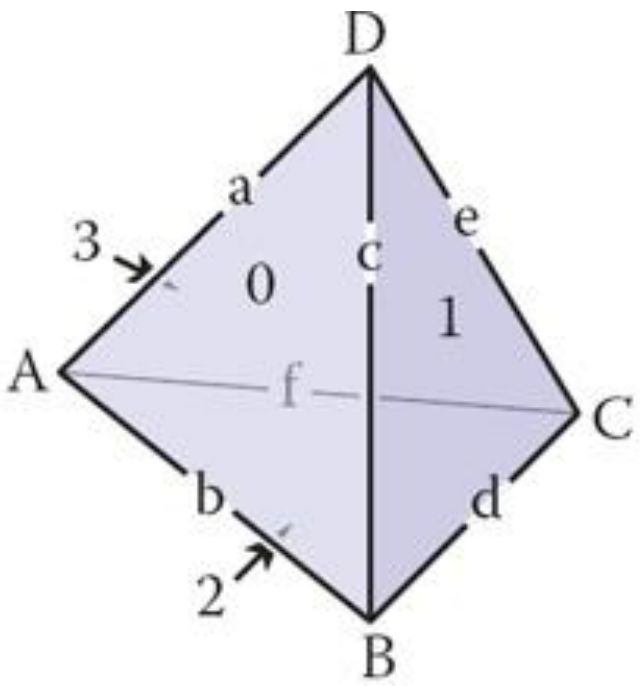
	v_1	v_2	v_3	v_4	v_5	v_6
v_1		1	1			
v_2	1		1	1		
v_3	1	1		1		1
v_4		1	1		1	1
v_5				1		1
v_6			1	1	1	

More about Winged-Edge

- Undirected edges
- The orientation of mesh decides which vertex is **head** and **tail**
- Requires consistent checking of orientation



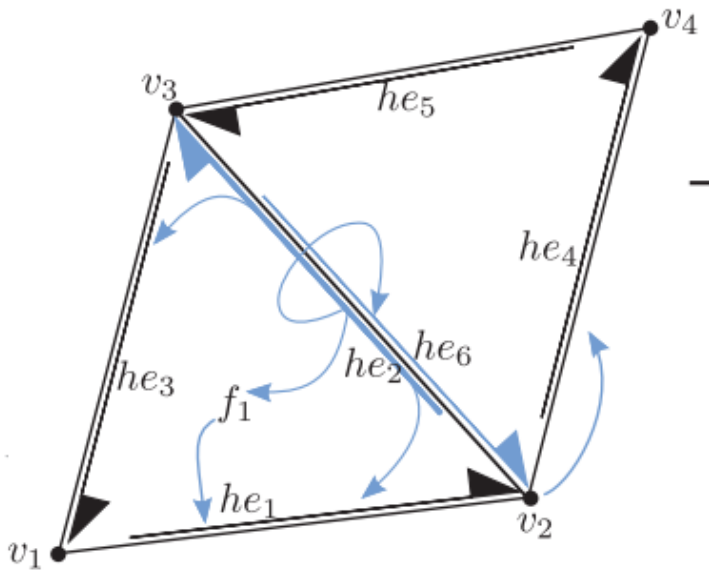
Winged Edge



Edge	*Vertex 1*	*Vertex 2*	*Face left*	*Face right*	*Pred left*	*Succ left*	*Pred right*	*Succ right*
a	A	D	3	0	f	e	c	b
b	A	B	0	2	a	c	d	f
c	B	D	0	1	b	a	e	d
d	B	C	1	2	c	e	f	b
e	C	D	1	3	d	c	a	f
f	C	A	3	2	e	e	b	d

More about Half-Edge

- Half-edge:
  - Directed half-edges
  - No consistent checking of which face the edge belongs to

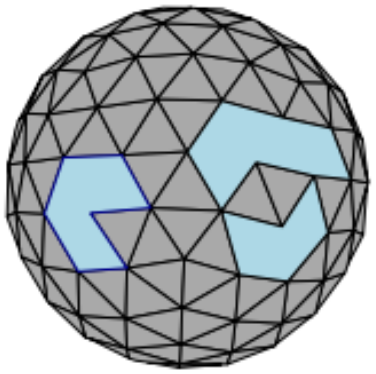


he_2		
$vert$	:	v_2
$prev$	:	he_1
$next$	:	he_3
opp	:	he_6
$face$	:	f_1

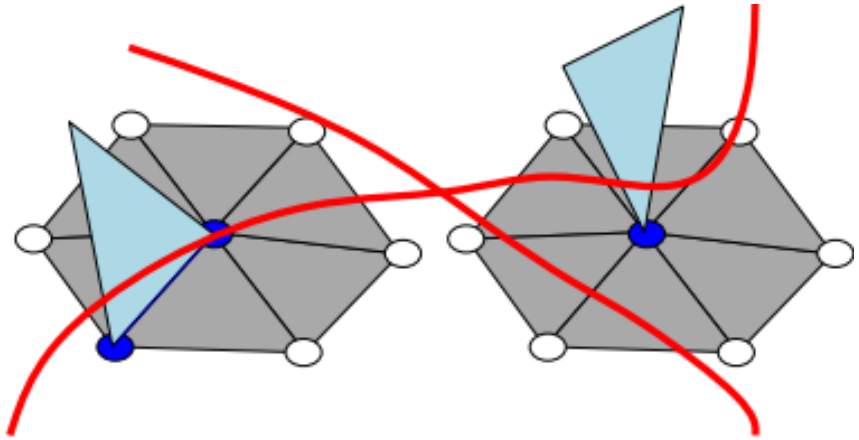
v_2		
$hedge$	:	he_4

f_1		
$hedge$	:	he_1

Edge-based Data Structures: Problems



- What kind of surfaces can be represented?
  - Meshes with holes? Tricky but doable
  - Non-manifolds?
  - Inconsistent orientation?
- Better for modification of meshes compared to face sets or indexed meshes



More Data Structures…

- Quad-edge
- Directed Edge
- Doubly Connected Edge List (DCEL)
- Corner tables – almost everything in $O(1)$ time!
- Face-based Data Structures

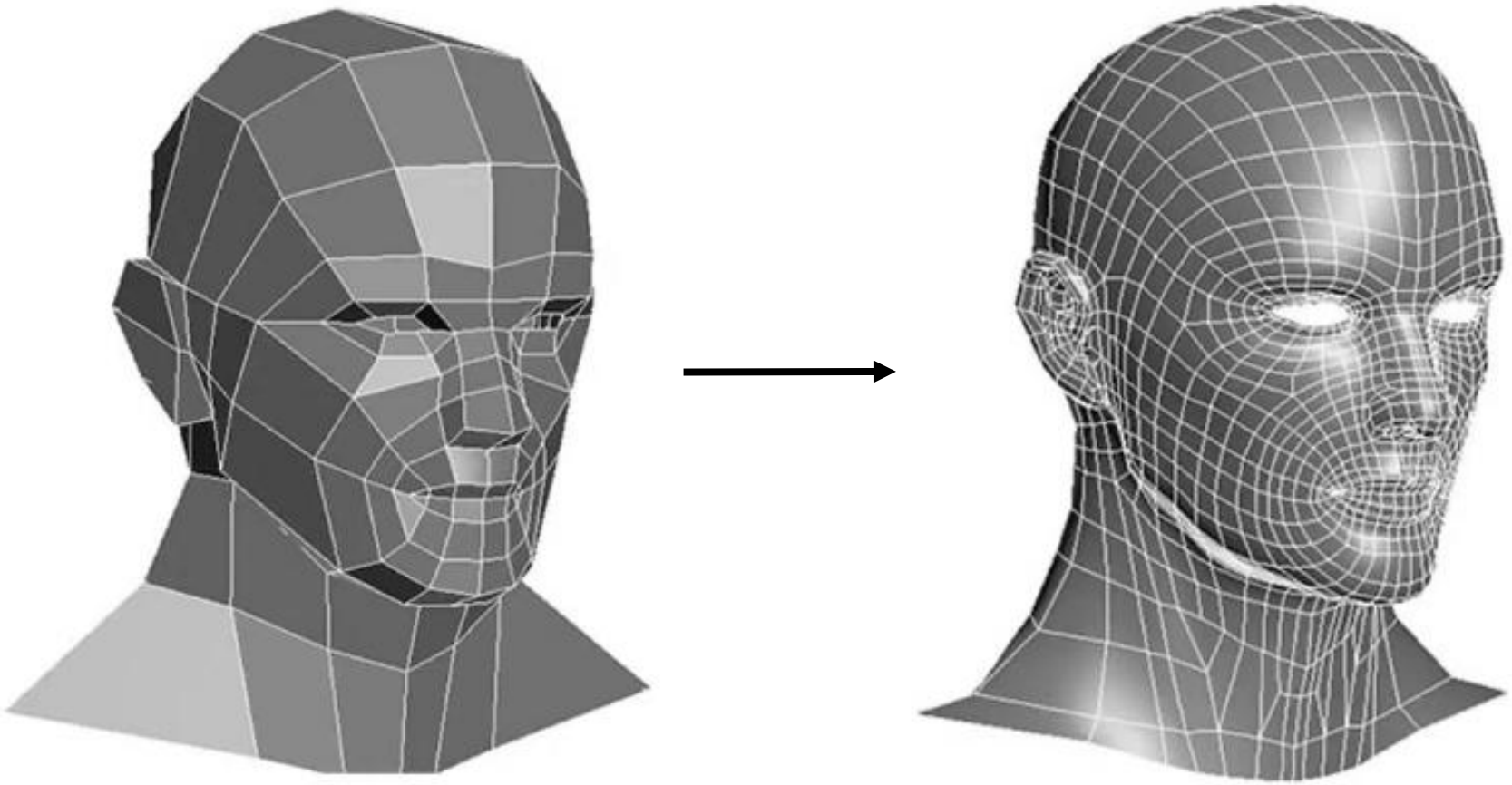
Mesh Data Structures Exercise!

Mesh Analysis and Operations

- Modification of 3D meshes
  - Subdivision
  - Smoothing

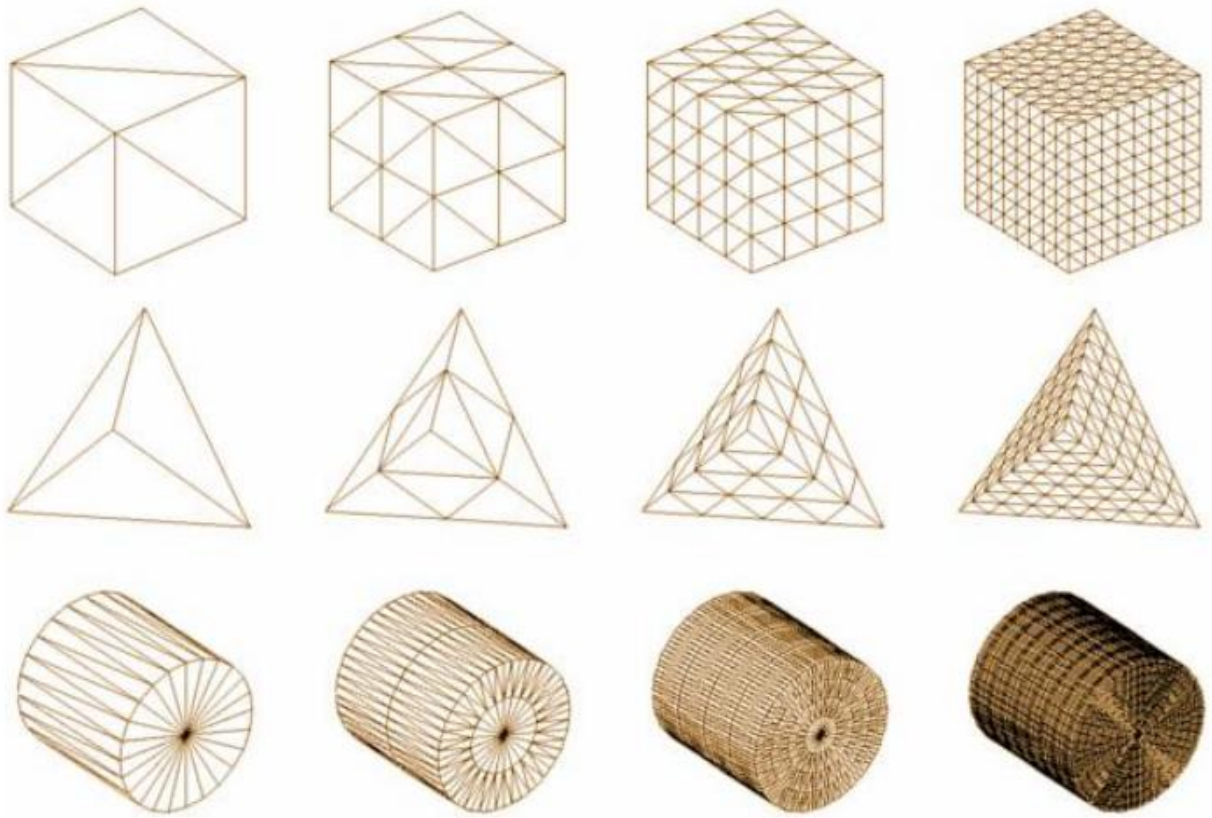
    Our focus for today!
  - Triangulation
  - Edge collapse (Removal of triangles)
  - Vertex Split (Addition of vertices)
- These algorithms make the graphics subfield: **geometry processing**
- Several packages/libraries support them natively
  - CGAL
  - OpenMesh

Subdivision



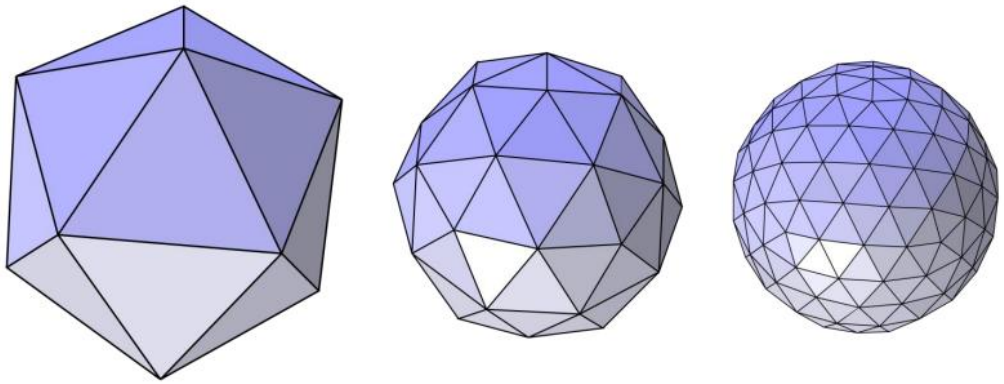
Linear Subdivision

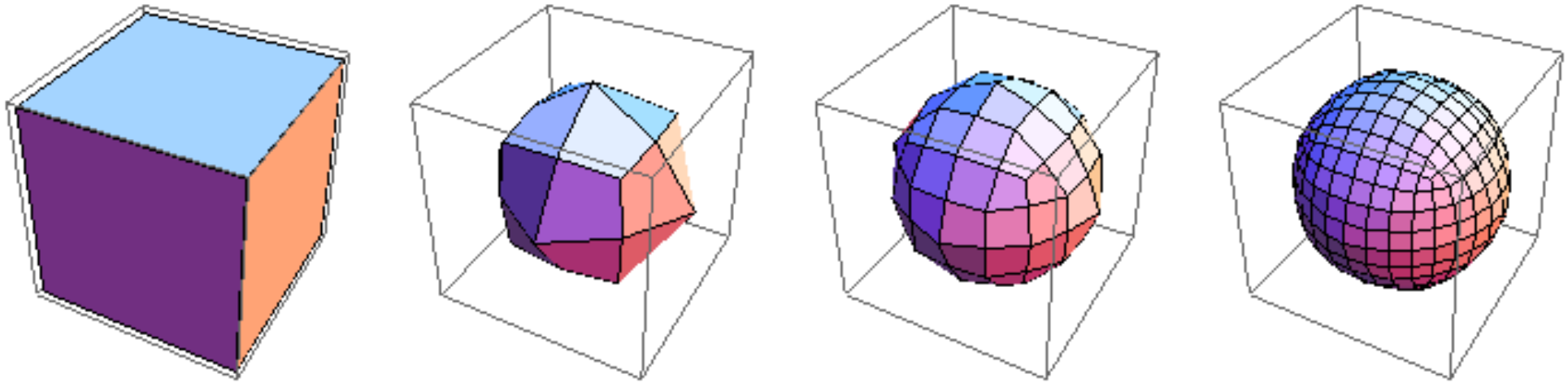
- Insert vertices at edge midpoints
- Adds faces, but doesn't change shape



Loop Subdivision

- Works for triangular meshes
- Algorithm:
  - Split each triangle into four triangles
  - Move the new vertices (using the weighted average of vertex neighbors)
  - Move old vertices based on the weighted average of its position and its neighbor's positions (valence)
  - Make smaller triangles



Catmull-Clark Subdivision

- Mostly works on quad meshes
- How it works: https://youtu.be/2UjfxAuXuSU?si=qePilYVBlBb1VaMD